

# Placement Portal Project Report

## Student Details

**Name:** Samyak Paul

**Roll Number:** 24f3004482

**Department:** BS in Data Science

## Project Details

### Problem Statement

The objective of this project is to develop a "Placement Portal" which is a centralized application for the placement process of campus. Generally traditional placements are coordinated by manual tracking of company registrations, application of students and job posting, where there is scope for human error and inefficiency, whereas this portal targets to be a bridge between companies, administrators and students and also provides a role based interface to manage the whole recruitment process.

### Approach

The problem was solved by building a multi-role web application using the Flask framework. The architecture focuses on three main entities:

**Administrators:** Controls the whole recruitment cycle from application to selection including approving companies, placement drives and also monitoring and rejecting them

**Companies:** Register their profiles, create placement drives with specific eligibility criteria. Also responsible for shortlist, select and reject student applications.

**Students:** Can apply, upload their resume and track their application through the portal.

### AI/LLM Declaration

I declare that AI/LLM tools (such as Gemini) were used for the model definition, and improving variable naming consistently. The extension of AI usage is around 15%. All final implementation logic and debugging were done manually.

### Frameworks and Libraries Used

Flask (2.3.3): Core web framework.

Flask-Login (0.6.3): User session and authentication management.

Flask-SQLAlchemy (3.0.5): Database abstraction Layer.

SQLAlchemy (1.4.46): SQL Toolkit and ORM.

Werkzeug (2.3.7): Security utilities for password hashing.

python-dotenv (1.0.0): Environment variable management.

Jinja2: Server-side HTML templating.

### ER Diagram

erDiagram

```
USER ||--o| COMPANY_PROFILE : "has"
USER ||--o{ PLACEMENT_DRIVE : "creates"
USER ||--o{ APPLICATION : "submits"
PLACEMENT_DRIVE ||--o{ APPLICATION : "receives"
```

```
USER {
    int id PK
    string email
    string name
    string role "admin/company/student"
    string contact
    string approval_state
    boolean is_blacklisted
    string student_roll
    string resume_filename
}
```

```
COMPANY_PROFILE {
    int id PK
    int user_id FK
    string company_name
    string hr_contact
    string website
    text description
}
```

```
PLACEMENT_DRIVE {
    int id PK
    int company_id FK
    string job_title
    text job_description
    string eligibility
    string deadline
    string status
    datetime created_at
}
```

```
APPLICATION {
    int id PK
    int student_id FK
    int drive_id FK
    string status
}
```

```
    datetime date_applied
}
```

### Table Relations:

**User & CompanyProfile:** One-to-One relationship (Role: Company).

**User & PlacementDrive:** One-to-Many relationship (A company creates multiple drives).

**User & Application:** One-to-Many relationship (A student submits multiple applications).

**PlacementDrive & Application:** One-to-Many relationship (A drive receives many applications).

## API Resource Endpoints

### Core Authentication

`/login` (GET/POST): User authentication.

`/register/<role>` (GET/POST): Role-specific registration (student/company).

`/logout` (GET): Session termination.

### Administrator Resources

`/admin/dashboard`: Overlook system statistics.

`/admin/companies`: Manage company approvals and blacklists.

`/admin/students`: Manage student database and access.

`/admin/drives`: Approval workflow for new job postings.

`/admin/applications`: Global view of all recruitment activities.

### Company Resources

`/company/dashboard`: View drive performance and applicant count.

`/company/profile`: Manage corporate identity and contact details.

`/company/drives/new`: Post a new placement opportunity.

`/company/drives/<id>/edit`: Update job details or eligibility.

`/company/applications/<id>/status`: Update student application state.

### Student Resources

`/student/dashboard`: View all "Approved" placement drives.

`/student/drive/<id>`: Detailed view and application submission.

`/student/profile`: Resume upload and information management.

`/student/history`: Personal application tracking.